

# FPGA-Accelerated ECG Filtering and Peak Analysis

Experiment 3 (Intermediate) | V-SPACE FPGA Build Challenge 2026

| Field             | Value                                                                                      |
|-------------------|--------------------------------------------------------------------------------------------|
| Platform          | PYNQ-Z2 / Zynq XC7Z020 (xc7z020clg400-1)                                                   |
| Tools             | Vitis HLS / Vivado 2025.1.1; PYNQ Python; independent Icarus Verilog 12.0 RTL verification |
| Confirmed members | Shanshank Pulipati - 25BEC0573; Kartik Yadav - 25BEC0087                                   |
| Team name         | Beyond Boolean                                                                             |

## 1. Objective

Accelerate 128-tap, 16-bit Q4.12 FIR preprocessing in FPGA programmable logic, transport samples through AXI DMA and verify the numerical output against an independent fixed-point reference. Demonstrate candidate ECG peak detection and heart-rate estimation in Python on the ARM processor using recorded MIT-BIH data. Live ECG sensor acquisition and clinical arrhythmia classification are outside the demonstrated scope.

## 2. Block Diagram

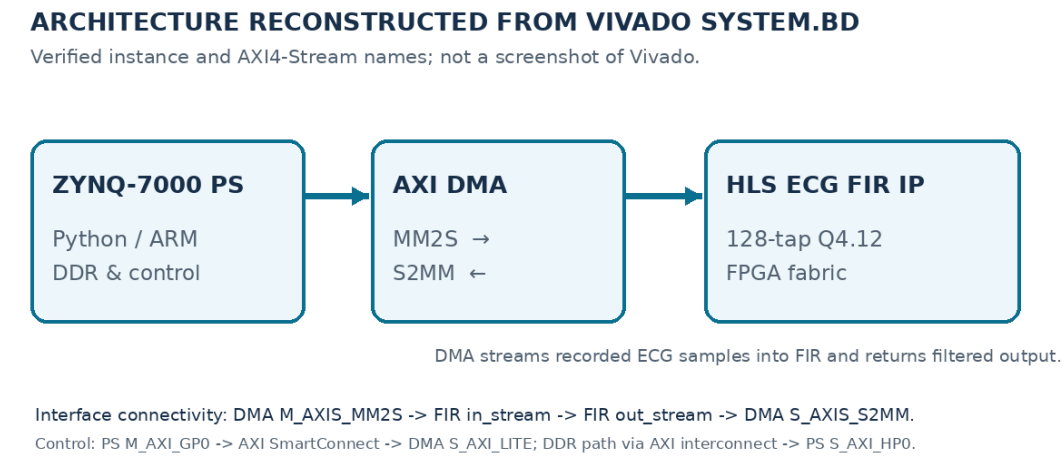


Figure 1. Architecture reconstructed from verified system.bd net names. This is a source-derived block diagram, not a Vivado GUI capture.

DDR samples pass through AXI DMA MM2S, the HLS FIR in\_stream/out\_stream interfaces and DMA S2MM back to memory. The Zynq PS controls DMA over AXI-Lite and accesses memory through the high-performance interconnect path.

### 3. RTL Design and Module Hierarchy

| Module / source                                 | Responsibility                                                                     |
|-------------------------------------------------|------------------------------------------------------------------------------------|
| ecg_bandpass_filter.cpp / .h                    | Original HLS C++ FIR function, fixed-point types and stream interface              |
| ecg_bandpass_filter.v                           | Actual generated FIR top, count=2048 in the board design, active-low control reset |
| Pipeline_VITIS_LOOP_79_1 and mac_muladd modules | Generated pipelined filtering and multiply-add dependencies                        |
| regslice_both                                   | Generated stream register slices                                                   |
| system_wrapper.v / system.bd                    | Vivado PS, DMA, interconnect and FIR integration                                   |

The expanded original IP package and all generated Verilog dependencies are supplied. The Vivado wrapper requires its block-design/vendor-IP hierarchy; the full original workspaces remain linked in SOURCE\_PROVENANCE.md. Supplied XDC files are generated IP/out-of-context constraints rather than a claimed custom board pin constraint file.

### 4. Simulation Results

Original Vitis HLS C-simulation logs show a pass. Independent Icarus Verilog 12.0 simulation on 9 October 2026 now additionally checks the unchanged generated RTL. Four stimulus groups include impulse, sine, seeded signed random values and recorded ECG. Input gaps, output stalls, repeated 2048-sample batches, KEEP/STRB preservation and TLAST are tested.

| Verification                    | Observed result                                                   |
|---------------------------------|-------------------------------------------------------------------|
| Independent RTL numerical check | 28,672/28,672 exact samples; maximum absolute error 0 LSB         |
| Stimulus composition            | 20,480 payload + 8,192 zero-flush samples                         |
| Batches / TLAST                 | 14 batches; all frame and invocation counts pass                  |
| Stream checks                   | Output stability during backpressure; KEEP/STRB and TLAST pass    |
| Reference cross-check           | ECG integer reference agrees with saved physical-board output CSV |

The generated FIR static state is cleared between independent inputs with a full zero batch, matching the existing board notebook. The testbench waits for ap\_done and pipeline state writeback. The report does not assume that control reset alone clears FIR history.

### 3. RTL Design - Actual Vivado Capture

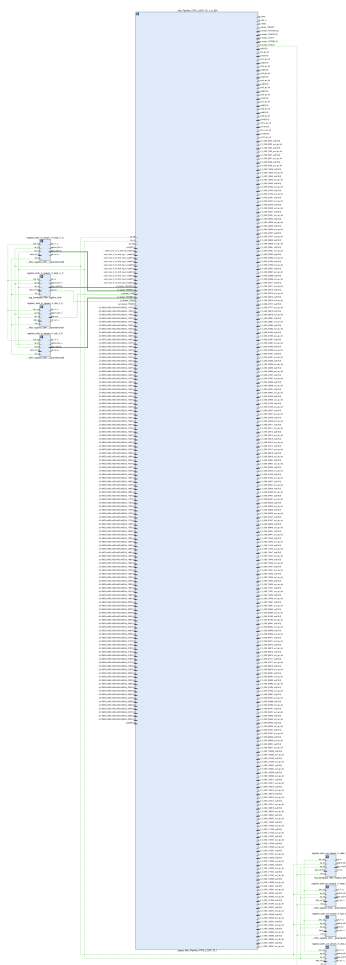


Figure 2. Actual Vivado 2025.1.1 elaborated FIR hierarchy exported by the team: one generated filtering pipeline and eight AXI-stream register-slice instances (9 cells, 28 displayed nets). The full vector PDF is supplied as *Images/rtl\_schematic.pdf* for detailed zooming. This is a selected module hierarchy view; top-level primitive logic is retained in the source files.

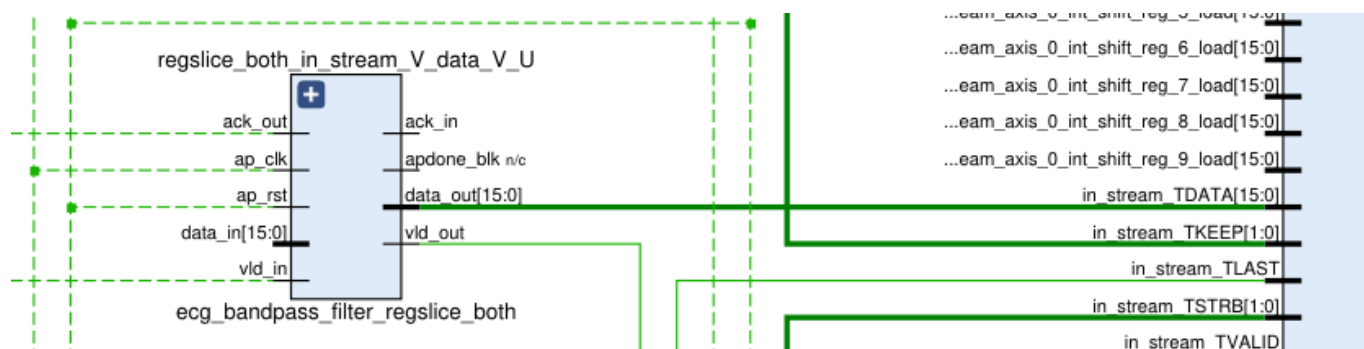


Figure 3. Magnified input-register-slice excerpt rendered directly from the same unmodified Vivado PDF. Labels and connections are preserved.

## 4. Simulation Results - Actual Waveform

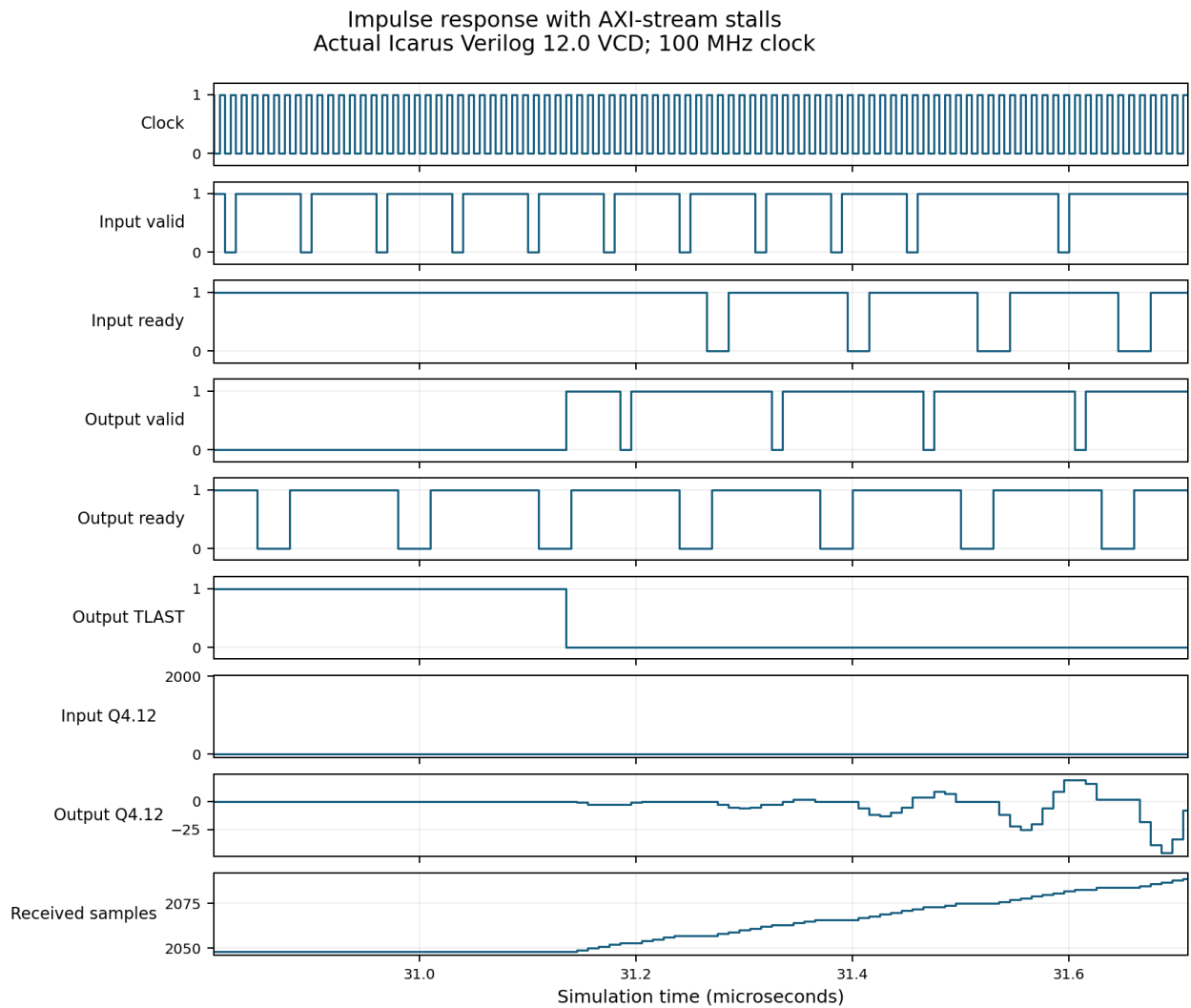


Figure 4. Actual generated-RTL waveform decoded from the saved Icarus VCD. All waveform points come from the executed simulation; this is not a conceptual timing drawing.

The full run is preserved in `Simulation/rtl_waveform.vcd.gz`. `Simulation/transcript.txt` contains the commands and actual passing outputs; `rtl_output_samples.csv` contains all 28,672 checked samples. The standalone simulation report provides the test scope and acceptance criteria. This unit simulation does not model the entire Zynq PS/DDR/DMA system.

## 5. Hardware Implementation

The original physical PYNQ-Z2 demonstration loaded the overlay and detected axi\_dma\_0. The matching BIT and HWH are retained unchanged. The XSA contains the same bitstream. The screenshot below is a recovered original Jupyter capture showing the loaded overlay and available IP; it is not a fresh board run.

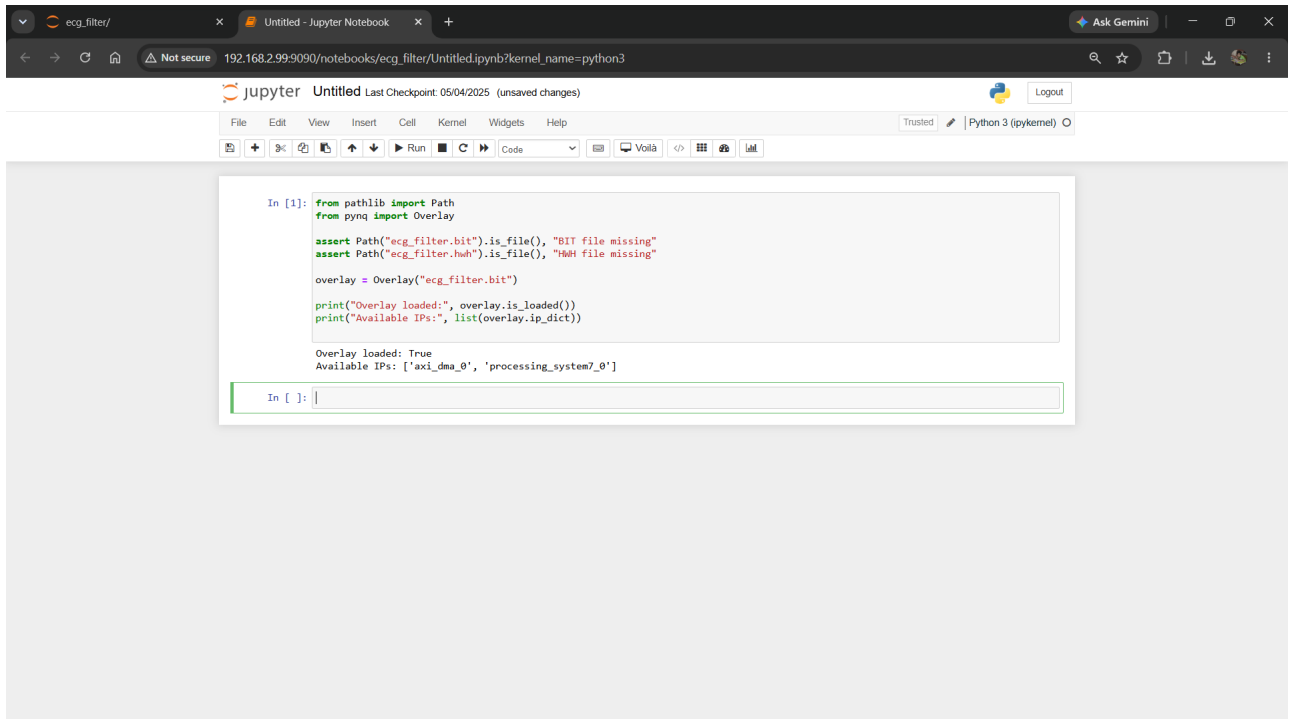


Figure 5. Actual user-supplied Jupyter screenshot: Overlay loaded: True; axi\_dma\_0 and processing\_system7\_0 detected. The original PNG is preserved; hardware\_output.jpg is its uncropped format derivative.

| Saved physical-board measurement | Result                                               |
|----------------------------------|------------------------------------------------------|
| Synthetic fixed-point comparison | 2,048/2,048 exact samples; 0 Q4.12 LSB maximum error |
| 5 Hz gain / 50 Hz reduction      | 0.9940 / 53.74 dB in the demonstrated sine test      |
| Recorded ECG comparison          | 12,288/12,288 exact; six 2048-sample data batches    |
| Python candidate peak analysis   | 42 peaks; mean RR about 0.812 s; heart rate 73.9 bpm |
| Python + DMA wall time           | 7.55 ms in the saved run; not FPGA-only latency      |

Whole-project placed resources from the original Vivado report: 7,232 slice LUTs, 13,844 slice registers, 2 block RAM tiles and 40 DSPs. These are not FIR-only counts. The original timing and routed DRC reports remain supplied.

## 5. Hardware Implementation - Saved Output Plots

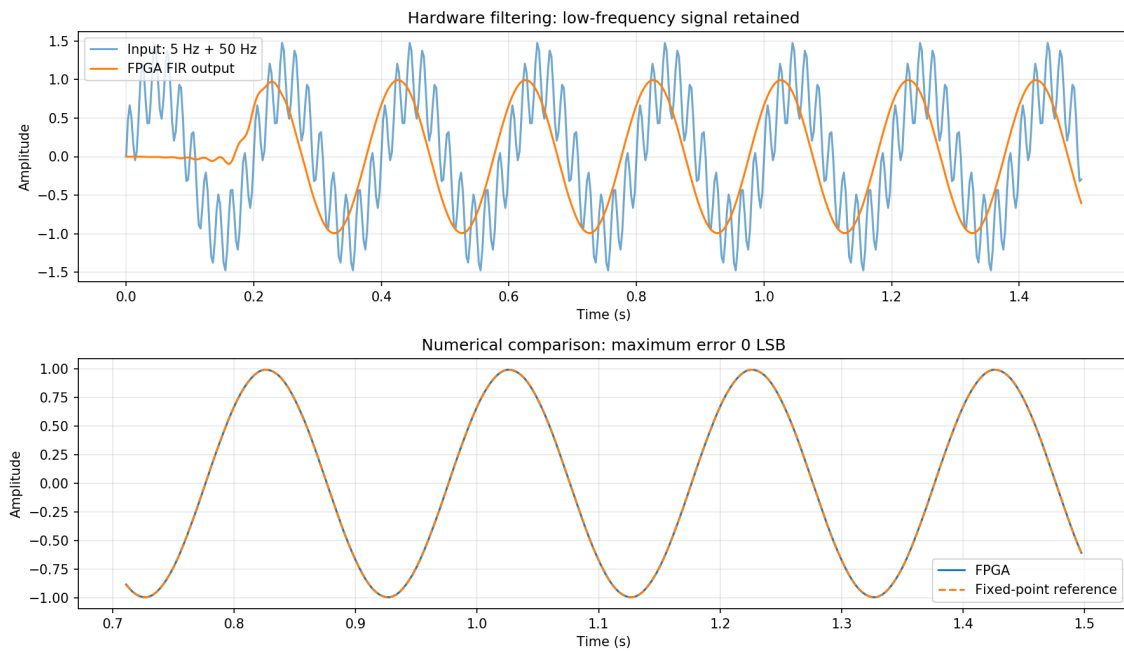


Figure 6. Original saved board sine-response and fixed-point comparison plot.

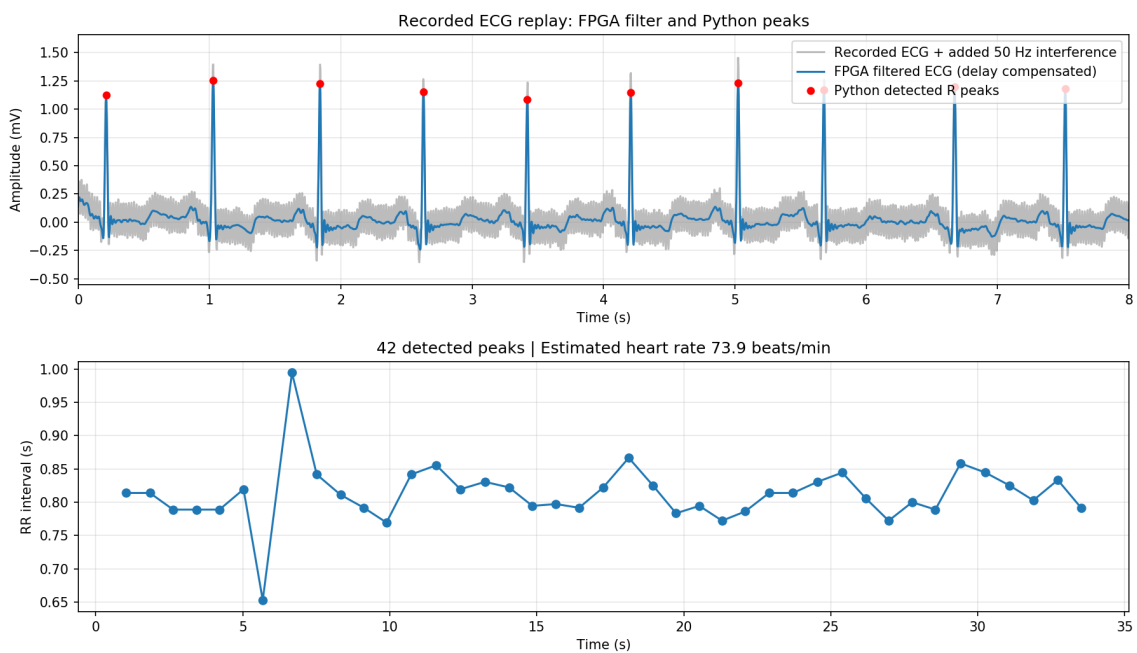


Figure 7. Original saved board ECG replay and Python peak analysis. The record is MIT-BIH 100 MLII, sampled at 360 Hz, with 0.15 mV added 50 Hz interference. FIR filtering is FPGA logic; peak detection is Python on ARM.

The actual setup photograph and Vivado capture have now been supplied and are incorporated in this report.

## 5. Hardware Implementation - Physical Setup



Figure 8. Actual team-supplied PYNQ-Z2 setup photograph, supplied 9 October 2026. The USB and Ethernet connections and illuminated board LEDs are visible. The original JPEG bytes are retained as `Images/board_setup.jpg`. The photo documents the setup; numerical hardware-test evidence remains in the saved executed notebooks and result files.

## 6. Applications

Educational PS/PL biomedical signal processing; FPGA preprocessing for embedded signal-analysis systems; and demonstrations of fixed-point accuracy, memory-to-stream DMA and hardware/software co-design. Medical diagnosis would need separate sensor interfacing and validation; that capability is not claimed by this demonstration.

## 7. Conclusion and Future Scope

The saved physical-board demonstration verifies exact filtering for synthetic and recorded ECG inputs and performs candidate peak/heart-rate analysis on ARM. The additional independent RTL test verifies exact core samples and tested stream protocol behavior under stalls and across batch boundaries. Future work includes live acquisition, broader signal validation, hardware peak detection and controlled performance benchmarking.

## Evidence and Reproduction

Original HLS/Vivado workspaces, the original packaged IP and saved physical-board results are preserved or linked in SOURCE\_PROVENANCE.md. Testbench/run\_rtl\_sim.py reproduces the independent unit simulation. The waveform PNGs are derived from its actual VCD. rtl\_run\_metadata.json identifies the simulator and hashes the tested RTL/testbench files. FILE\_MANIFEST.json checksums the package.

MIT-BIH Arrhythmia Database v1.0.0, record 100 MLII, samples 0-12287, 360 Hz. Source: <https://physionet.org/content/mitdb/1.0.0/>. Attribution/license details are in PYNQ\_Hardware/DATA\_SOURCE.md.

## Submission Status

The Project 3 file set now includes the required four images, Verilog testbench, actual RTL waveform/transcript/simulation report, confirmed team details and this seven-section project report. Team: Beyond Boolean; Shanshank Pulipati (25BEC0573) and Kartik Yadav (25BEC0087). The final demo URL is the remaining Project 3 submission item; Video\_Link.txt stays blank as requested.

The seven-section format follows the organizer-provided documents\_req.pdf. The full five-project report is a separate team-level deliverable.